Experiment no : 3 : Subnetting.

→ aim : to write a use program to demo subnetting & find subnet masks.

→ objective :
- understand IPV4, IPV6
- understand address & subnet classes
- learn subnetting & subnet mask calc.
- determine network, host ID.

→ theory :

- ip :
  it helps us provide a unique address to devices & enables communication over a network.

IPV4 : · 32- bit
· in dotted decimal notation

IPV6 : · 128- bit
· in hexadecimal denotation

- classful addressing :

class	range	subnet
A	1 - 126	255.0.0.0
B	128 - 191	255. 255 . 0.0
C	192 - 223.	255. 255. 255. 0

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D	224 - 239	multicast
E	240 - 255	experimental

- network & no of hosts per subnet:

no of hosts:

hosts: $2^{\wedge}$(host bits) - 2

eg. 192.168.1.0/24

host bits = 8.

hosts ⇒ $2^8 - 2 = \boxed{254}$.

- subnetting:

it refers to divide the large network into smaller subnets by borrowing bits from the host portion of the IP.

eg. 192.168.1.0/24

subnet mask: 225.255.255.192

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⇒ FAQ:

1) classful addressing divides IPV4 address into fixed classes. Each class has a predefined network & host portion with a fixed default subnet mask.

advantages:
- simple to understand
- easy implementation

disadvantages:
- wastes IP address because # subnet size fixed.

class A: 10.0.0.1
class B: 172.16.5.10
class C: 192.162.1.20

2) private address:
- 10.0.0.0 /8
- 172.16.0.0 /12
- 192.168.0.0/16

loopback address:
- 127.0.0.0/8

broadcast address:
- 255.255.255.255

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3.

Subnetting is the process of dividing one large network into smaller networks to improve performance & efficiently use IP addresses.

4.

FLSM (fixed length subnet mask) : all subnets have same subnet mask.

VLSM (variable length subnet mask) :

different subnet masks are used based on host requirements.

CIDR (classless inter domain routing) :

uses slash notation for efficient IP allocation & routing.

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